

Pre-Practicum 3

(EBN 111 : Electricity and Electronics)

Introduction to: Operational Amplifier Circuits

EBN 111: Practical 3

Agenda



- ☐ Objectives
- ☐ Description of Operational Amplifiers
- ☐ Physical Construction on Breadboard
- ☐ Physical connections of the Operational Amplifier circuit
- ☐ Practical Preparation

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Objectives



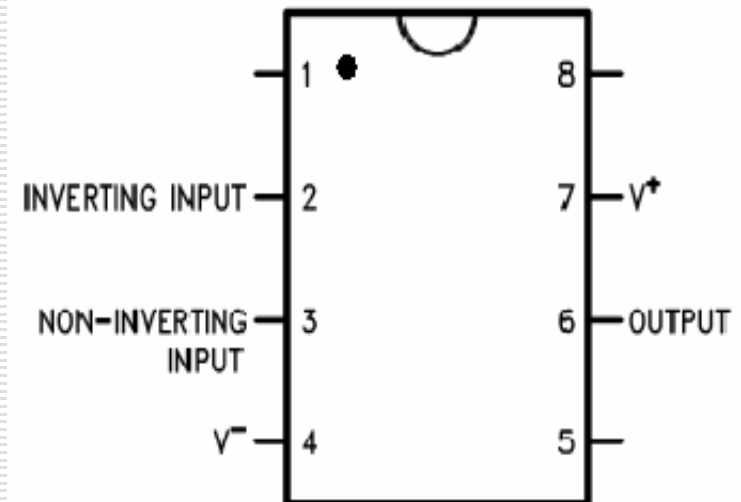
- ❑ Build two Operational Amplifier (Op-Amp) circuits
- ❑ Then record the Op-Amp transfer characteristics
 - **i.e.** Input Voltage versus Output Voltage
- ❑ **Important:** Read about the basic concepts of Inverting and Non-Inverting amplifiers in your textbook
- ❑ These components will be provided during the practical
 - Resistors
 - UA741 Operational Amplifier
 - Connecting wires

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Description of Op-Amps

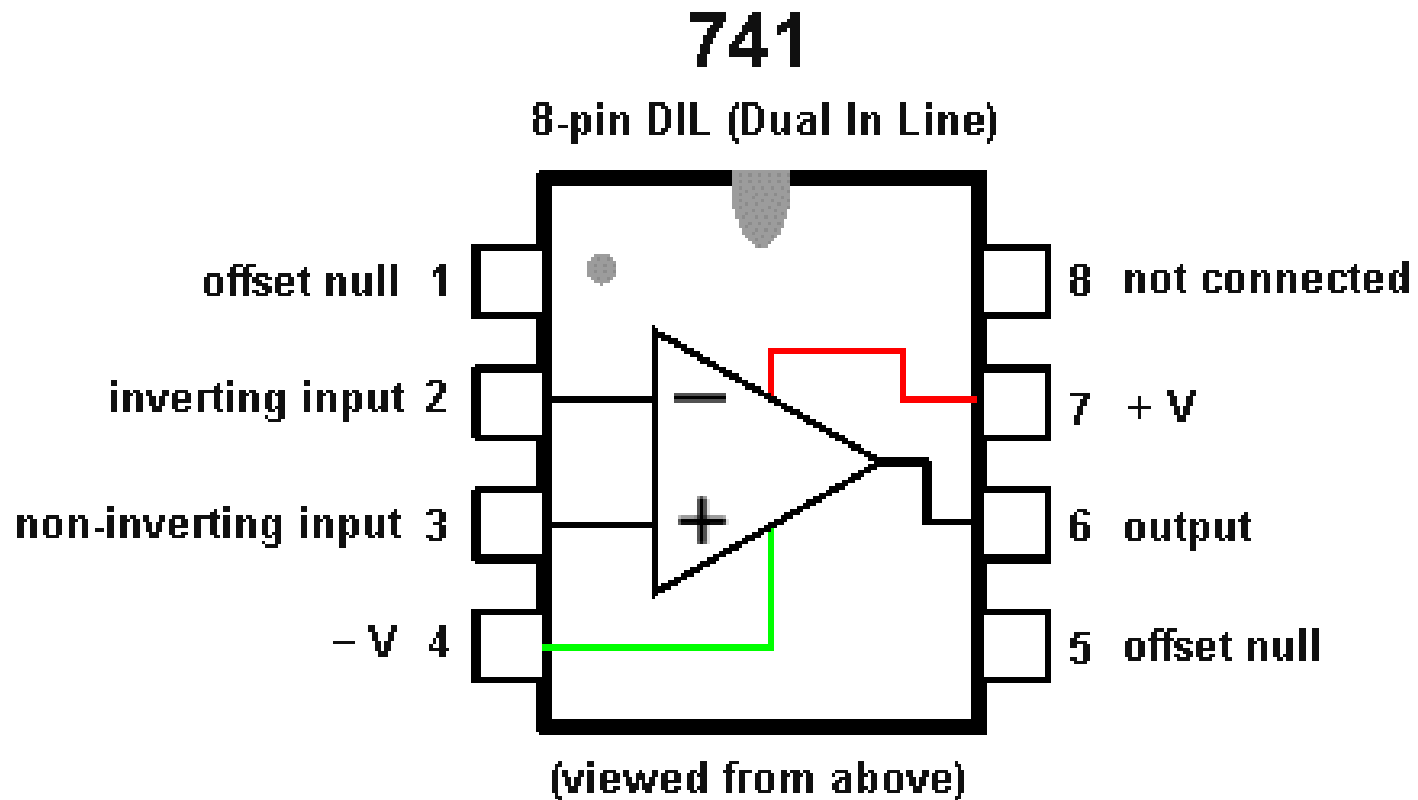


- ❑ UA741 Op-Amp will be used for this practical, it will be in 8 pin Dual In-Line Package
- ❑ The physical layout of the Op-Amp is shown here
- ❑ The pins correspond to the various inputs and outputs of the Op-Amp
- ❑ **Note:** Pin 1 is identified with the dot next to it and all other pins are counted counter-clockwise from pin 1



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Description of Op-Amps

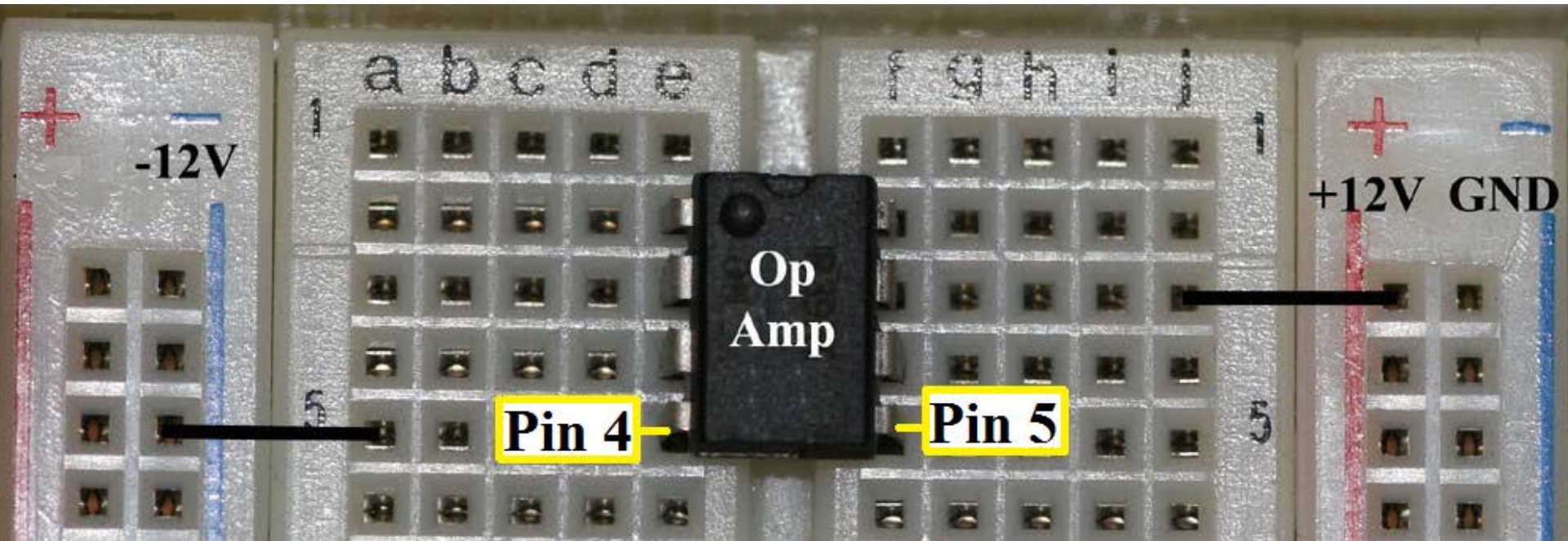


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Constructions on Breadboard



- Place the Op-Amp on the breadboard so that pin 4 and 5 are separated by the vertical column break on the board
- *Hint:* Connect the -12V, 12V and ground (COM) rails to the breadboard as indicated



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Agilent Workstation



Power Supply



Function generator (Oscilloscope)



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Physical Connections



- ❑ Connect V+ (pin 7) to the +12 V power supply
- ❑ Connect V- (pin 4) to the -12 V power supply
- ❑ Connect the input signal (either pin 2 or 3) to the function generator
- ❑ All grounds must be connected to the ground (COM)
- ❑ Connect resistors according to specific question (1 or 2)
- ❑ Two circuits will be constructed:
 - Inverting Amplifier
 - Non-Inverting Amplifier

Read practical 3 guidelines for more detail

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Practical Preparation



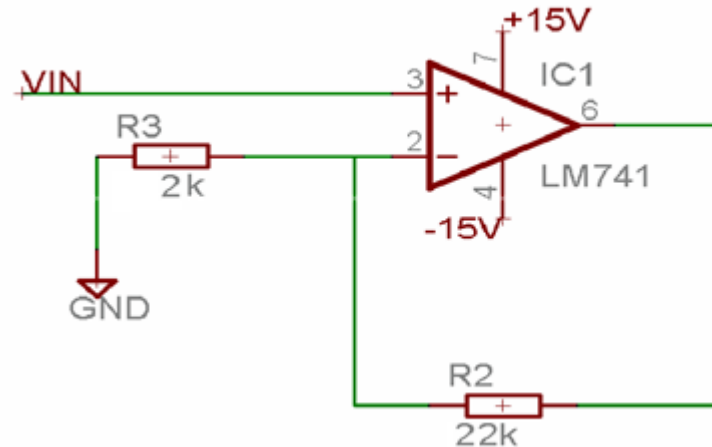
- ❑ **Important:** Closed shoes are required in the labs
- ❑ Practical 3 dates are, **21 -24 October**
- ❑ Practical 3 guidelines are available on ClickUp
- ❑ Read through practical guidelines and familiarize yourself with the components and testing sequence
- ❑ Bring your **component boxes**

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Practical Preparation



- Prepare for the practical test by doing the following questions and figure below:
 - Is the output inverted from the input?
 - What is the theoretical gain of the amplifier?
 - What is the relationship between input and output voltage?
 - Amplification and saturation regions

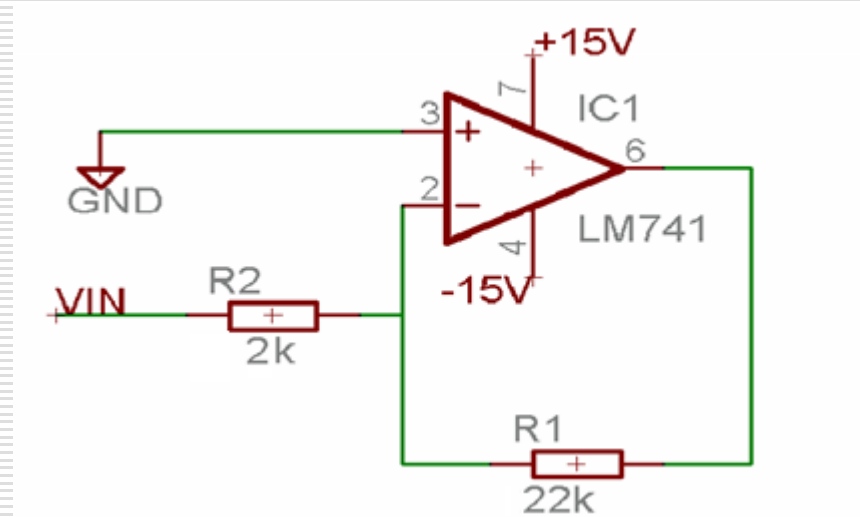


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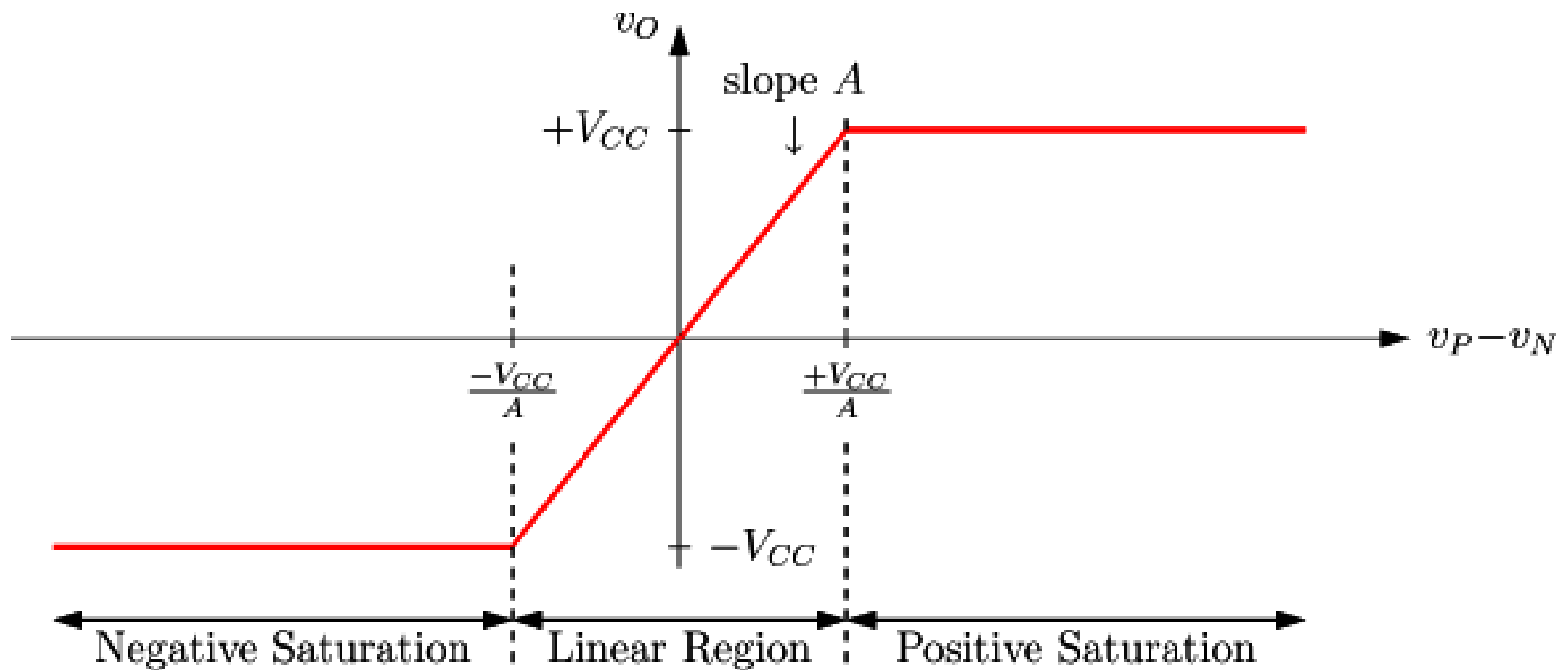


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Practical Preparation



- Amplification and saturation regions



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Conclusion



- ❑ Handle the Op-Amp with care
- ❑ Questions can be emailed to Rogerio dos Santos
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